In vitro and in vivo efficacy of sulfo-carrabiose, a sugar-based cosmetic ingredient with anti-cellulite properties.

Most of adult women exhibit cellulite on the hips, buttock and thighs. Although extracellular matrix and lymphatic system disorders can increase its appearance, cellulite basically results from an excessive fat storage in the adipose tissue which exerts considerable pressure on the surrounding skin tissue and creates a dimpled irregular appearance. Caffeine, the most widely used anti-cellulite ingredient, favours fat break-down by inhibiting the phosphodiesterase enzyme and encouraging a high intracellular level of cAMP. A series of studies has shown that spermine and spermidine, two ubiquitous polyamines, encouraged fat storage and slowed fat break-down in the adipose tissue. Besides, it was shown that heparan sulfate glycosaminoglycans had a strong affinity for polyamines. To design a new cosmetic ingredient with anti-cellulite properties, we used molecular modelling to screen several ingredients with a structure similar to that of heparan sulfate glycosaminoglycans. This way, we identified sulfo-carrabiose as a potent molecule for trapping spermine and spermidine. These virtual results were first confirmed in tubo where sulfo-carrabiose was shown to dose-dependently inactivate spermine and spermidine. In vitro, adipocytes cultured with sulfo-carrabiose exhibited a significant reduction of lipogenesis and a significant increase of lipolysis. When sulfo-carrabiose was incorporated in a cosmetic formula, significant improvements were observed in thigh circumference, with better results than those obtained with caffeine after 28 days of use. Furthermore, a combination of caffeine and sulfo-carrabiose led to results significantly better than those obtained with caffeine alone. As measured by fringe projection, thigh volume was also significantly reduced after sulfo-carrabiose treatment. Finally, the appearance of cellulite assessed by clinical evaluation was also significantly reduced within 28 days.